

Simran Garg

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🌐 LinkedIn | 🐙 GitHub | 🚀 LeetCode

SUMMARY

- Motivated Computer Science undergraduate with expertise in Data Structures Algorithms, Machine Learning, and Generative AI. Hands-on experience building LLM-powered applications using LangChain and LangGraph, machine learning and deep learning models. Proficient in FastAPI, Docker, and API development for deploying scalable, production-ready AI applications.

EDUCATION

Ajay Kumar Garg Engineering College

B.Tech in Computer Science

CGPA: 8.9

Ghaziabad, Uttar Pradesh

2024 – 2028

Intermediate (Class 12)

Percentage: 96%

PROJECTS

- **Financial analyst using LangGraph** Live Project
 - "Built a production-grade multi-agent financial analyst using LangGraph — 7-node stateful pipeline with conditional routing, self-critique reflection loop, and MemorySaver checkpointing; implemented Pydantic v2 structured output for typed financial data extraction and dual ingestion (PDF filings via pdfplumber/PyMuPDF + live web data via Tavily/SEC EDGAR). Designed an LLM-as-critic peer review node with a 6-dimension scoring rubric that automatically routes weak analyses back for revision — same adversarial pattern used in Constitutional AI. Shipped a full Streamlit UI with 4 input modes, session state, analysis history, and report download; 25 pytest tests; Docker + Render deployment."
- **Clinical Decision Support RAG system**
 - Built a self-critiquing Clinical RAG system (Self-RAG + Corrective RAG + LangGraph) over 2,202 indexed clinical guideline chunks with hybrid retrieval (HyDE + BM25 + RRF + cross-encoder re-ranking) parallelized relevance checks reduced LLM call overhead by 5x. Deployed on aws with Qdrant Cloud backend; evaluated with RAGAS (faithfulness, context precision, answer relevancy, context recall).
- **LSTM-based Movie Dialogue Generator** Live Project
 - Developed an LSTM-based Movie Dialogue Generator using Python and TensorFlow/Keras to generate context-aware movie dialogues through next-word prediction. Implemented temperature-based sampling and top-k filtering for more natural and diverse text generation, along with data preprocessing and sequence modeling. Built an interactive Streamlit web application and deployed it on Render for real-time dialogue generation.

SKILLS

- **Programming Languages:** Python, Java, C
- **Machine Learning:** Supervised Learning, Unsupervised Learning, Feature Engineering, Feature Scaling, Model Evaluation, Cross Validation, Ensemble Learning, XGBoost, Classification, Regression, Clustering
- **Deep Learning:** ANN, CNN, RNN, LSTM, Transformer Architecture, Transfer Learning
- **Natural Language Processing:** Text Preprocessing, Tokenization, Word Embeddings, Sentiment Analysis
- **Generative AI LLMs:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, AI Agents, Prompt Engineering, Tool Calling, LLM Application Development
- **Frameworks Deployment:** FastAPI, Docker, Streamlit
- **Libraries Frameworks:** NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Keras, NLTK, SciPy
- **Problem-Solving:** Data Structures, Algorithms, Competitive Programming (LeetCode)

ACHIEVEMENTS

- Solved 200+ problems on LeetCode, focusing on Data Structures and Algorithms.

CERTIFICATIONS

- **Advanced RAG Certification-CampusX | Skilled in retrieval pipeline design, hybrid search, query rewriting, and RAG evaluation.** [Certificate Link](#)
- **ICAT (Internship Common Aptitude Test) – Demonstrated analytical and problem-solving skills.** [Certificate Link](#)
- **Completed Data Science Machine Learning — Udemy | Covered supervised/unsupervised learning, feature engineering, model evaluation, and end-to-end ML pipelines in Python** [Certificate Link](#)